

Wildcards in Linux

Wildcards are special symbols used to match file names or groups of files.

1. Asterisk *

Definition: Matches zero or more characters.

Usage/Formula:

```
command *
```

Examples:

```
ls *.txt
```

Shows all `.txt` files.

```
rm file*
```

Deletes files starting with file.

```
cp *.jpg Pictures/
```

Copies all `.jpg` files to Pictures folder.

2. Question Mark ?

Definition: Matches exactly one character.

Usage/Formula:

```
command ?
```

Examples:

```
ls file?.txt
```

Matches `file1.txt`, `fileA.txt`

```
rm data?
```

Matches `data1`, `data2`

```
cp ?.doc Docs/
```

Matches one-letter filenames like `a.doc`

3. Brackets []

Definition: Matches one character from a list or range.

Usage/Formula:

```
command [characters]
```

Examples:

```
ls file[123].txt
```

Matches `file1.txt`, `file2.txt`, `file3.txt`

```
rm test[a-c]
```

Matches `testa`, `testb`, `testc`

```
cp report[0-9].txt Backup/
```

Matches `report1.txt`, `report5.txt`

4. Negated Brackets [!]

Definition: Matches any one character NOT listed.

Usage/Formula:

```
command [!characters]
```

Examples:

```
ls file[!1].txt
```

Matches all except `file1.txt`

```
rm data[!a-c]
```

Deletes files not ending in a, b, or c

```
cp test[!0-9].txt Folder/
```

Matches non-number endings.

Brace Expansion

Brace expansion creates many files or folders quickly.

Example 1: Create Multiple Folders

```
mkdir Week{1..5}
```

Creates:

Week1 Week2 Week3 Week4 Week5

Example 2: Create Entire Directory Structure

```
mkdir -p School/{Math,English,Science}/{Homework,Notes}
```

Creates:

```
├── Math/
│   ├── Homework
│   └── Notes
├── English/
│   ├── Homework
│   └── Notes
└── Science/
    ├── Homework
    └── Notes
```

Example 3: Create Numbered Files

```
touch file{1..3}.txt
```

Creates:

```
file2.txt
file3.txt``
```